Gudapati Deepak Joy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.93
SRI SHIRDI SAI VIDYANIKETAN EM SCHOOL KADIAM AP	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2023	86.00%
SRI NAND TECHNO SCHOOL (EM)	Class X (Board of Secondary Education , Andhra Pradesh)	2021	99.83%

SKILLS

Programming Languages: C, C++, Python

Web Technologies: HTML, CSS, JavaScript, React, Node.js, Express.js

Databases: MongoDB, MySQL

Libraries/Tools: NumPy, Matplotlib, SciPy, Jupyter Notebook, Streamlit , Scikit-learn, git, github, Solid Edge

PROJECTS

AI-Powered Adaptive IAM & SSO Gateway

Tech: Node.js, Express.js, React, MongoDB Atlas, JWT, Passport.js, OAuth 2.0, SAML 2.0, OIDC

- **Engineered a centralized SSO & IAM** gateway serving **5,000+ users** across institute-wide applications, unifying **Local, OAuth 2.0, SAML 2.0, and OIDC** authentication.
- **Integrated an ML-driven risk-scoring layer** to evaluate login velocity, **geolocation, and proxy/VPN** signals, dynamically applying **MFA** step-up verification based on session risk.
- **Architected a dual-token authentication** and session-security layer with **15-minute** access JWTs and **7-day** refresh tokens, transparent token renewal, context-aware session tracking, and automatic multi-session revocation upon validation failures.
- **Hardened the authentication infrastructure** with **automated key rotation**, CIDR-based **IP whitelisting**, and immutable audit logging across **13+ security event types** for monitoring and post-incident review.

TCP Socket-Based Quiz System with Network Protocol Design

Tech: C++, TCP Sockets, Multithreading, JSON, Custom Application Protocol, LLM Integration

- **Architected a multithreaded C++ TCP server** with a custom length-prefixed application protocol and **10-client connection backlog**, reliably handling message boundaries across quiz, answer, and leaderboard communication.
- **Implemented a custom KEEP-ALIVE/ALIVE_OK** heartbeat protocol with per-client threads and mutex-protected state, using **30-second heartbeats and 60-second** inactivity timeouts for automatic dead-client detection and connection cleanup.
- **Integrated an LLM-based quiz engine** generating **10 structured MCQs** per request, with JSON parsing and fallback handling for malformed model responses.
- Engineered a **thread-safe genre-level cache** to re-serve previously generated quizzes, eliminating redundant LLM API calls for repeated genre requests.

Radiomics MRI Analysis Web Application

Tech: Python, SimpleITK, PyRadiomics, Scikit-learn, Flask/React, DICOM/NIfTI, Multithreading

- **Built an end-to-end MRI preprocessing pipeline** covering denoising, N4 bias-field correction, normalization, coregistration, and skull-stripping, with a DICOM-to-NIfTI conversion module to standardize scans for analysis.

- **Applied radiomics and ML techniques** for tumor identification, achieving **~90% accuracy across 500 MRI scans**.
- **Engineered volumetric MRI processing** with N4 correction using **4 multi-resolution levels × 50 iterations** and 3D coregistration using **50 histogram bins and 100 optimization iterations**.
- **Implemented multithreaded scan processing** to offload compute-intensive operations and maintain responsive, non-blocking interaction during MRI analysis.

Dataset Distillation using LoRA and PEFT Methods

Tech: Machine Learning, Deep Learning, Dataset Distillation, Parameter-Efficient Fine-Tuning

- Conducted extensive research on **dataset distillation and PEFT** by integrating LoRA with dataset distillation paradigms on MobileNetV2 architectures.
- Implemented and evaluated dataset distillation frameworks including **meta-learning, gradient matching, trajectory matching (MTT), and SRe2L** statistical recovery on CIFAR-10 using custom LoRA-based convolutional layers.
- Developed a comprehensive experimental pipeline involving **LoRA integration, gradient-tracking ablations, synthetic dataset optimization, and cross-evaluation** analysis to study the compatibility of low-rank adaptation with dataset distillation techniques.
- Analyzed performance degradation and representational instability in **low-rank constrained models**, demonstrating limitations of combining LoRA with dataset distillation.

RELEVANT COURSES

Computer Science: Data Structures and Applications, Algorithms, Computer Networks, Software Engineering, Database Management Systems, Operating Systems, Theory of Computation, Discrete Mathematics.

Artificial Intelligence: Machine Learning, Fraud Analytics, AI for Finance

Mathematics: Convex Optimisation, Probability and Random Variables, Linear Algebra, Statistics, Calculus, Differential Equations, Transform Techniques, Complex Variables

EXTRACURRICULAR

- Active **volleyball player**, participating in multiple institute-level tournaments.

- Active **NSS participant**, contributing to walkathons, plantation drives, and social initiatives.